Diffusion Model for 3D ϕ^4 Theory

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Training Data

We prepare 10240 samples by solving the 3D ϕ^4 theory on a 32^3 lattice.

$$S_E = \sum_x \left[-2\kappa \sum_{\mu=1}^{d} \phi(x)\phi(x+\hat{\mu}) + (1-2\lambda)\phi^2(x) + \lambda\phi^4(x) \right] \tag{1}$$

where $\kappa = 0.5$, $\lambda = 2.5$ which is in the broken phase.

Samples are generated by the Langevin dynamics.

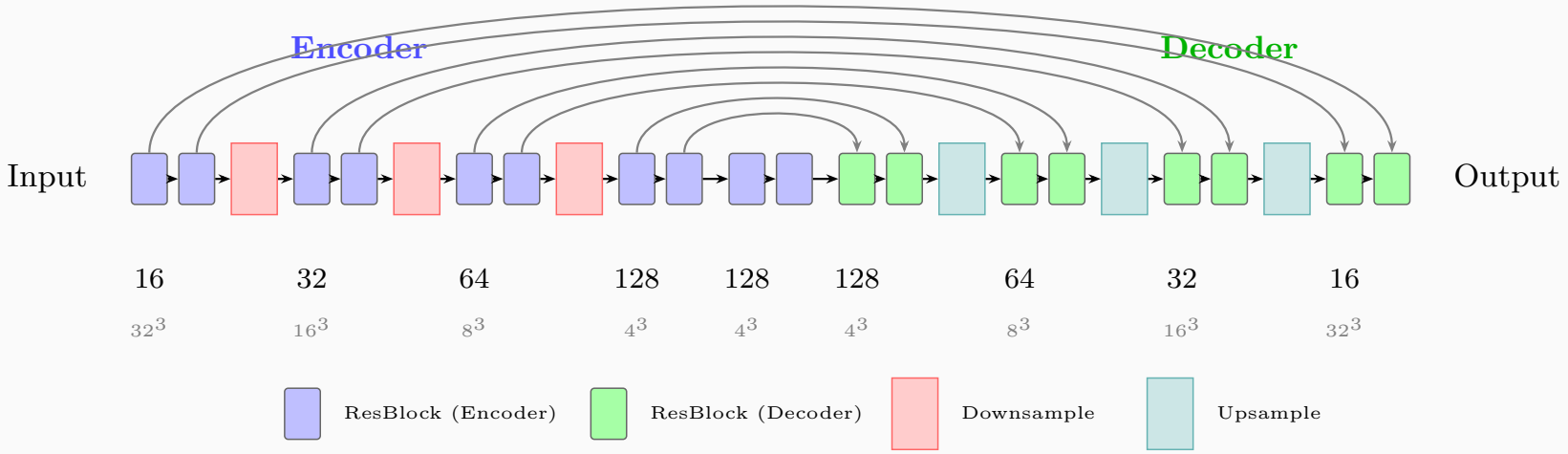
$$\frac{\partial \phi}{\partial t} = -\frac{\delta S_E}{\delta \phi} + \eta(x,t) \tag{2}$$

where $\langle \eta(x,t)\eta(x',t') \rangle = 2\delta(x-x')\delta(t-t')$.

We simulate the Langevin equation to time $t = 10$ to get the training data.

Model Architecture

We use the U-Net architecture for the diffusion model.



Some modifications are made to U-Net architecture in arxiv:2309.17082.

- We add two Residual blocks(without skip connections) between the CNN layers.
- We use the downsampling and upsampling layers to reduce the spatial resolution.
- $\sigma_{\text{max}} = 205$ is determined by the maximum distance between the training samples.

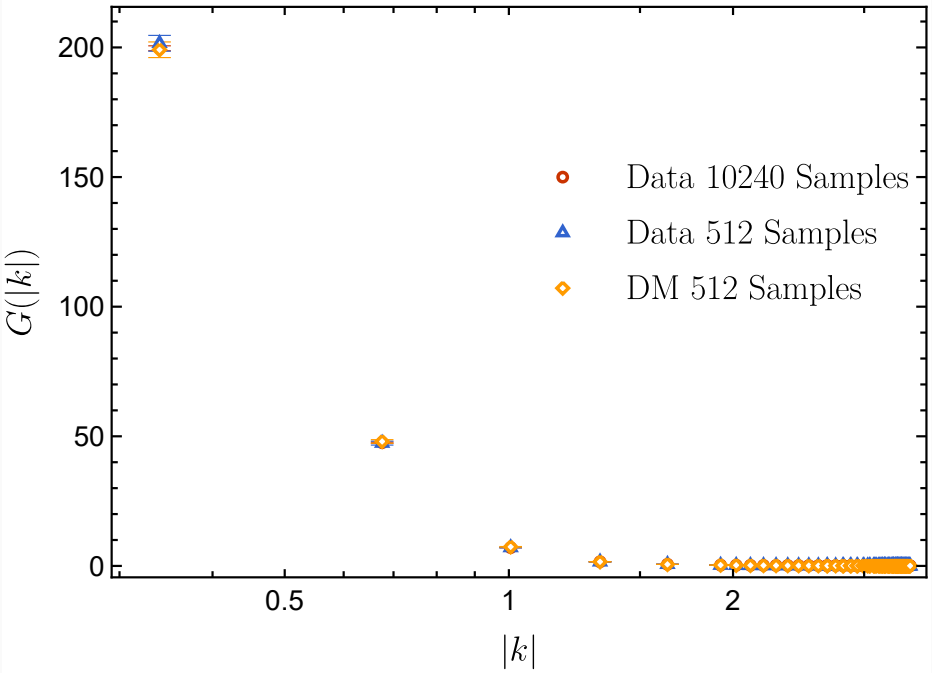
Extracting the propagator

In momentum space, using $\tilde{\phi}(-k) = \tilde{\phi}^*(k)$ for real ϕ:

$$G(k) = \langle \tilde{\phi}(k)\tilde{\phi}(-k) \rangle = \left\langle |\tilde{\phi}(k)|^2 \right\rangle, \quad \tilde{\phi}(k) = \frac{1}{\sqrt{V}} \sum_x e^{-ik\cdot x} \left[\phi(x) - \langle \phi \rangle\right] \tag{3}$$

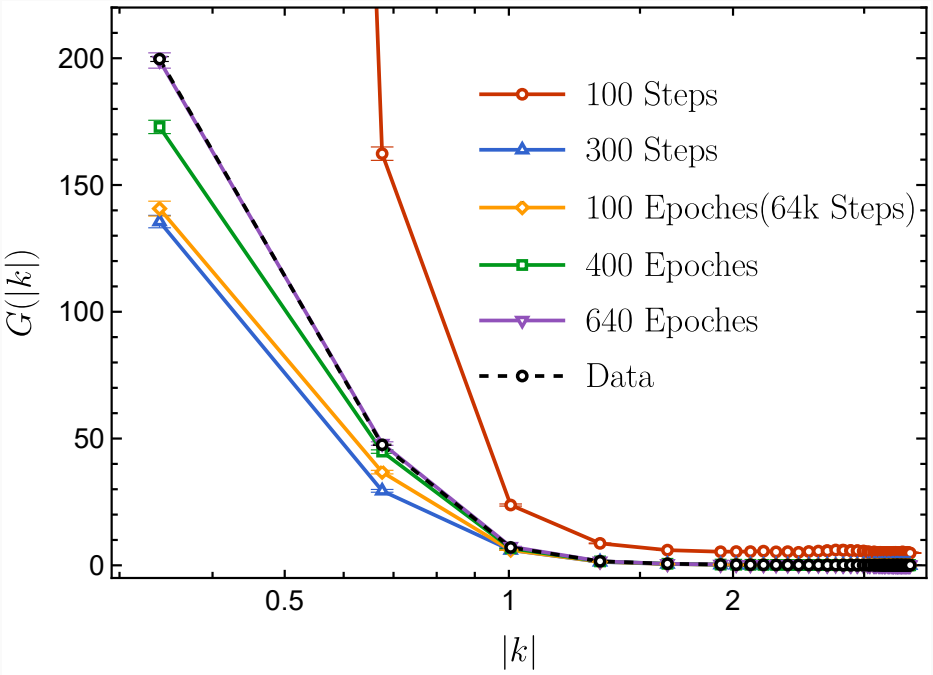
We use the lattice momentum to correct for discretization effects

Error bars are estimated using Jackknife resampling.

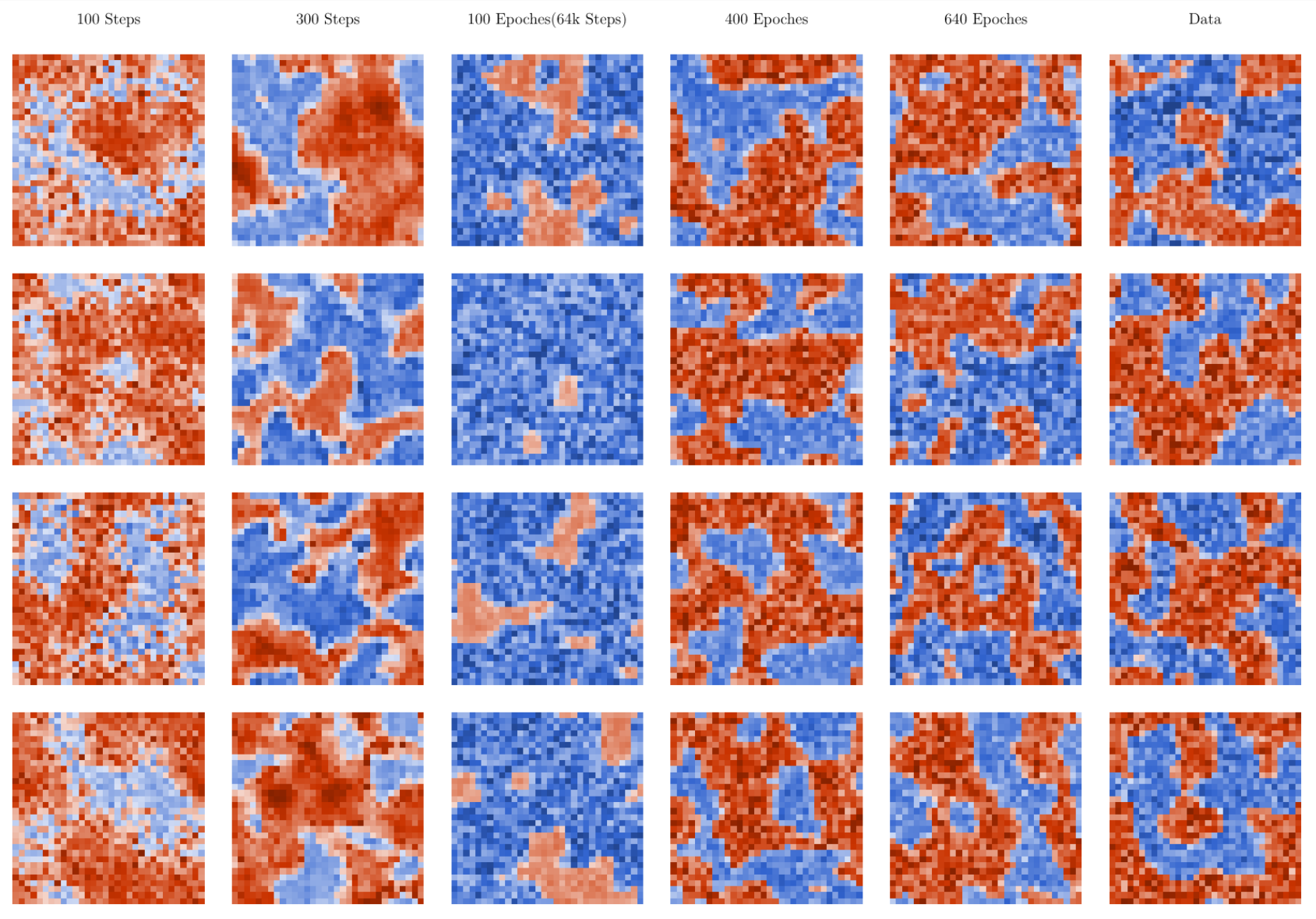


We select the most diagonal momentum mode.

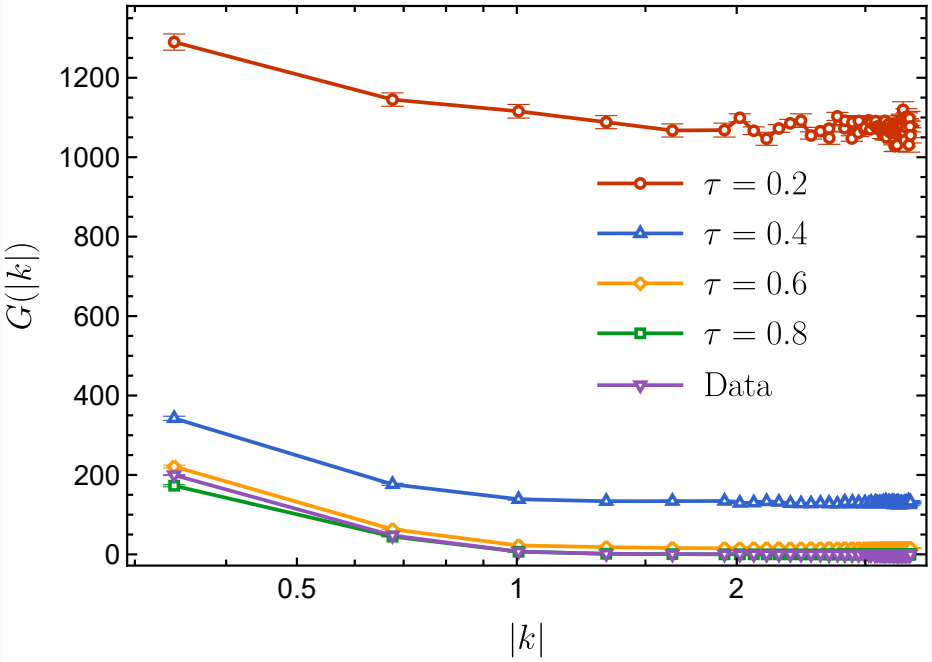
The evolution of the propagator during training



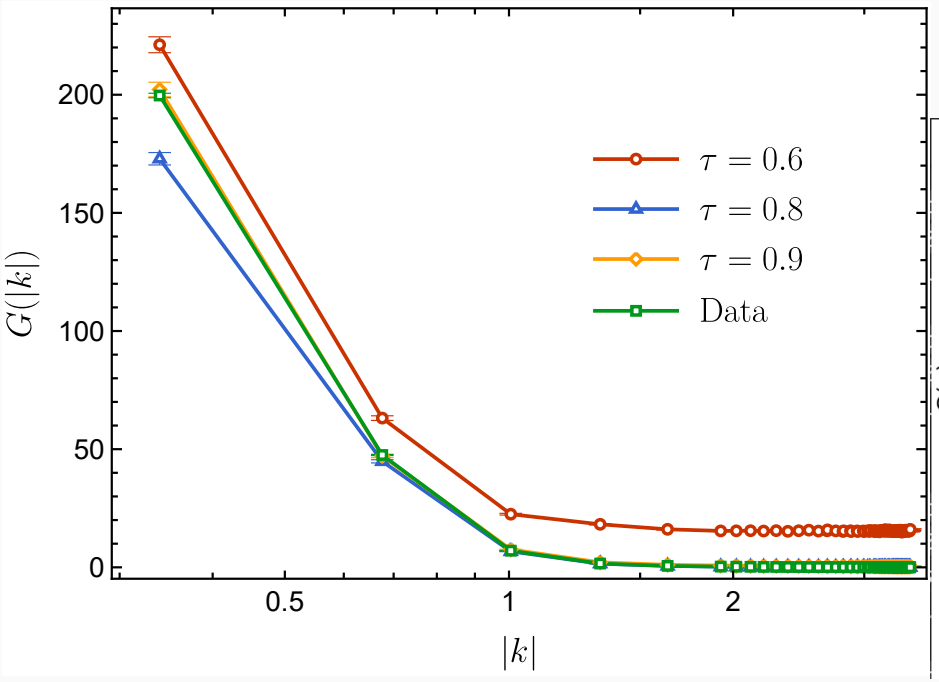
The evolution of the spatial pattern during training

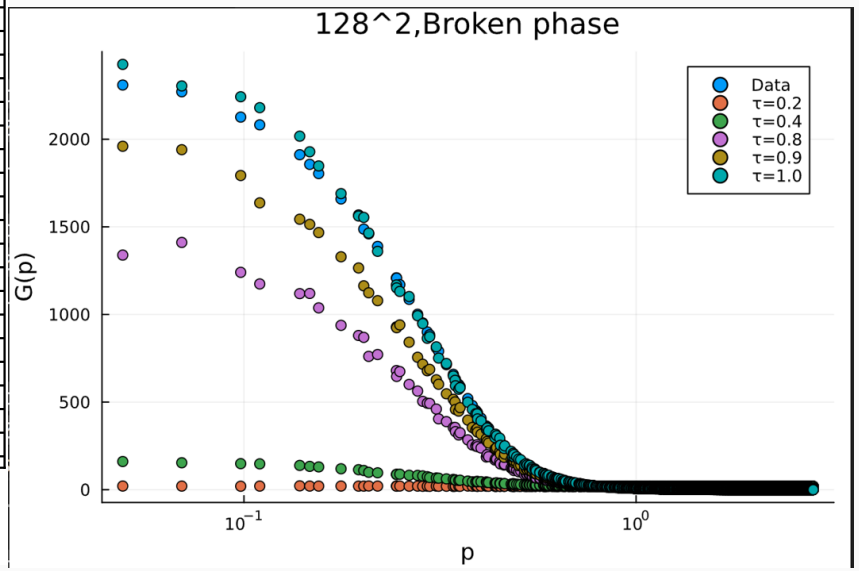


The evolution of propagator during backward diffusion



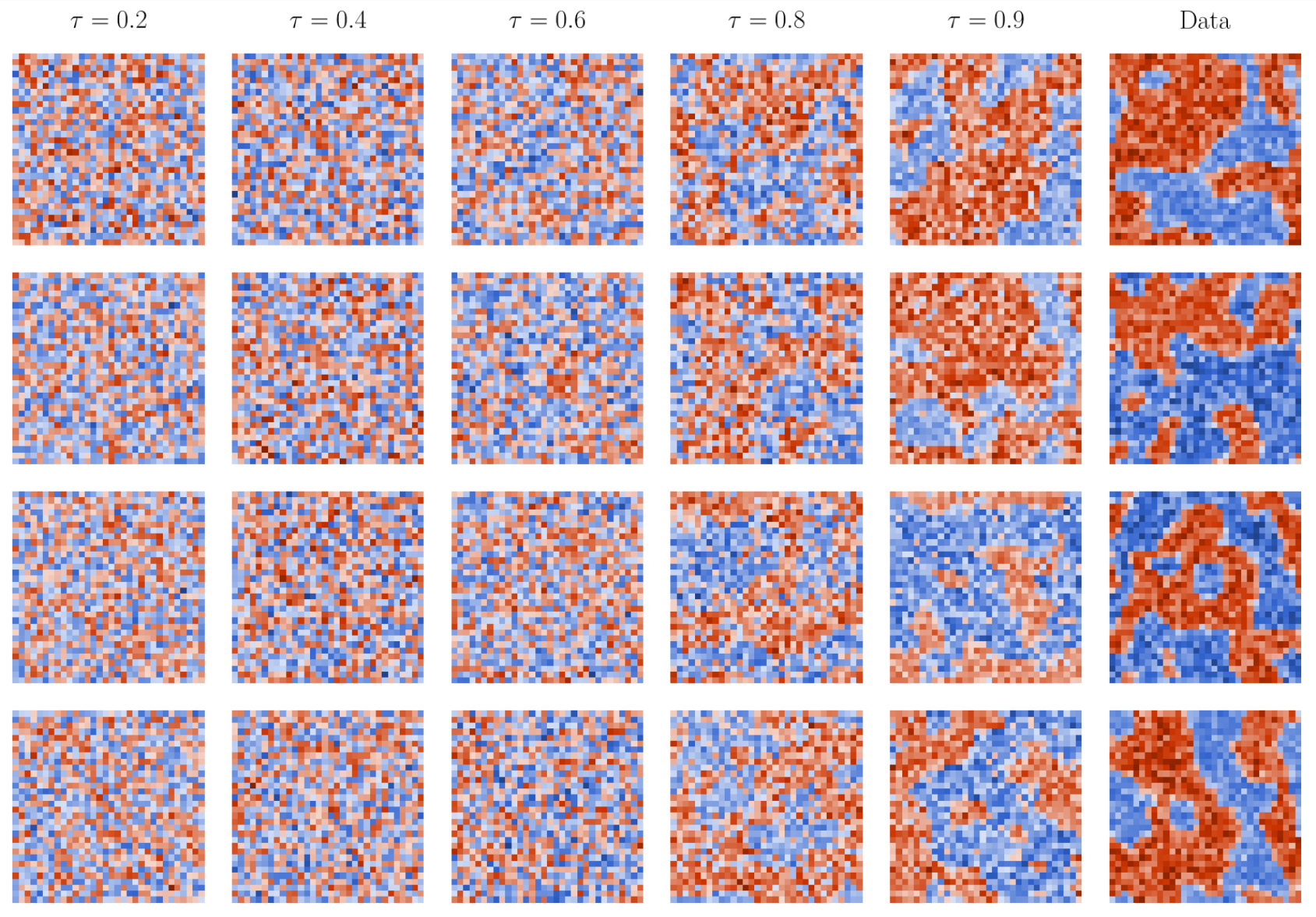
The evolution of propagator during backward diffusion





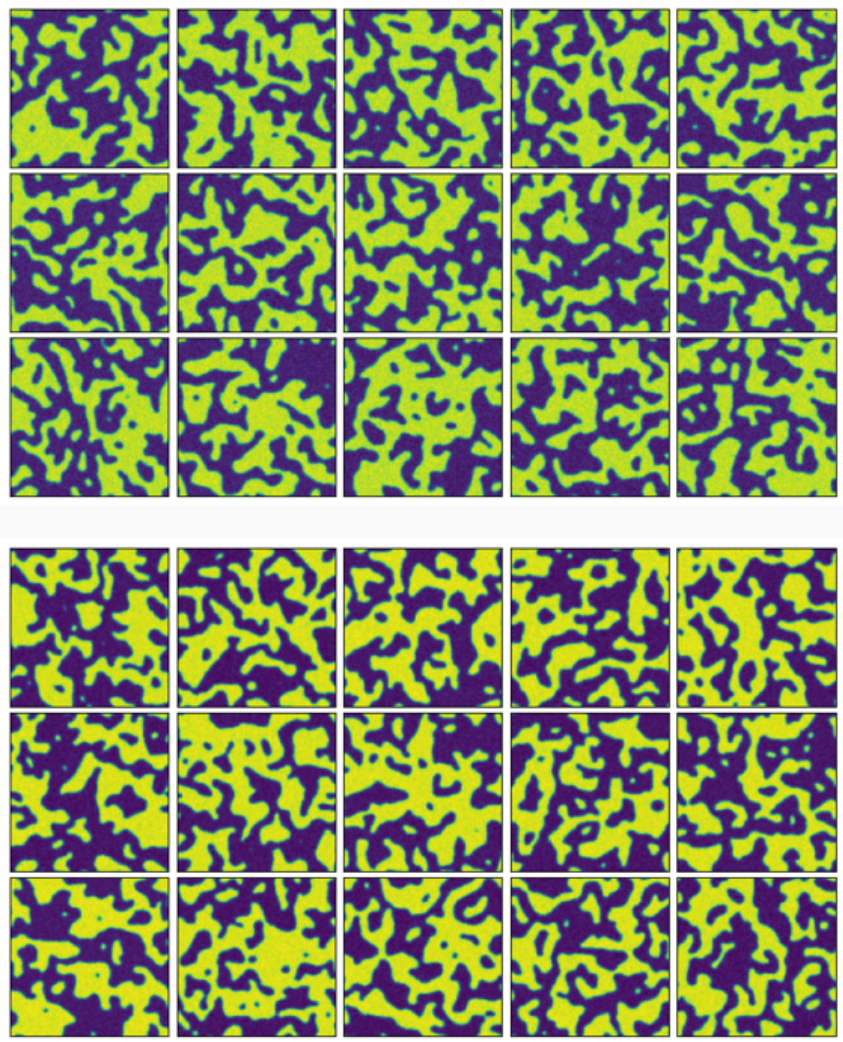
- Left: 3D propagator ($\sigma_{\text{max}} = 205$)
- Right: 2D propagator ($\sigma_{\text{max}} = 25$)

The evolution of spatial pattern during backward diffusion



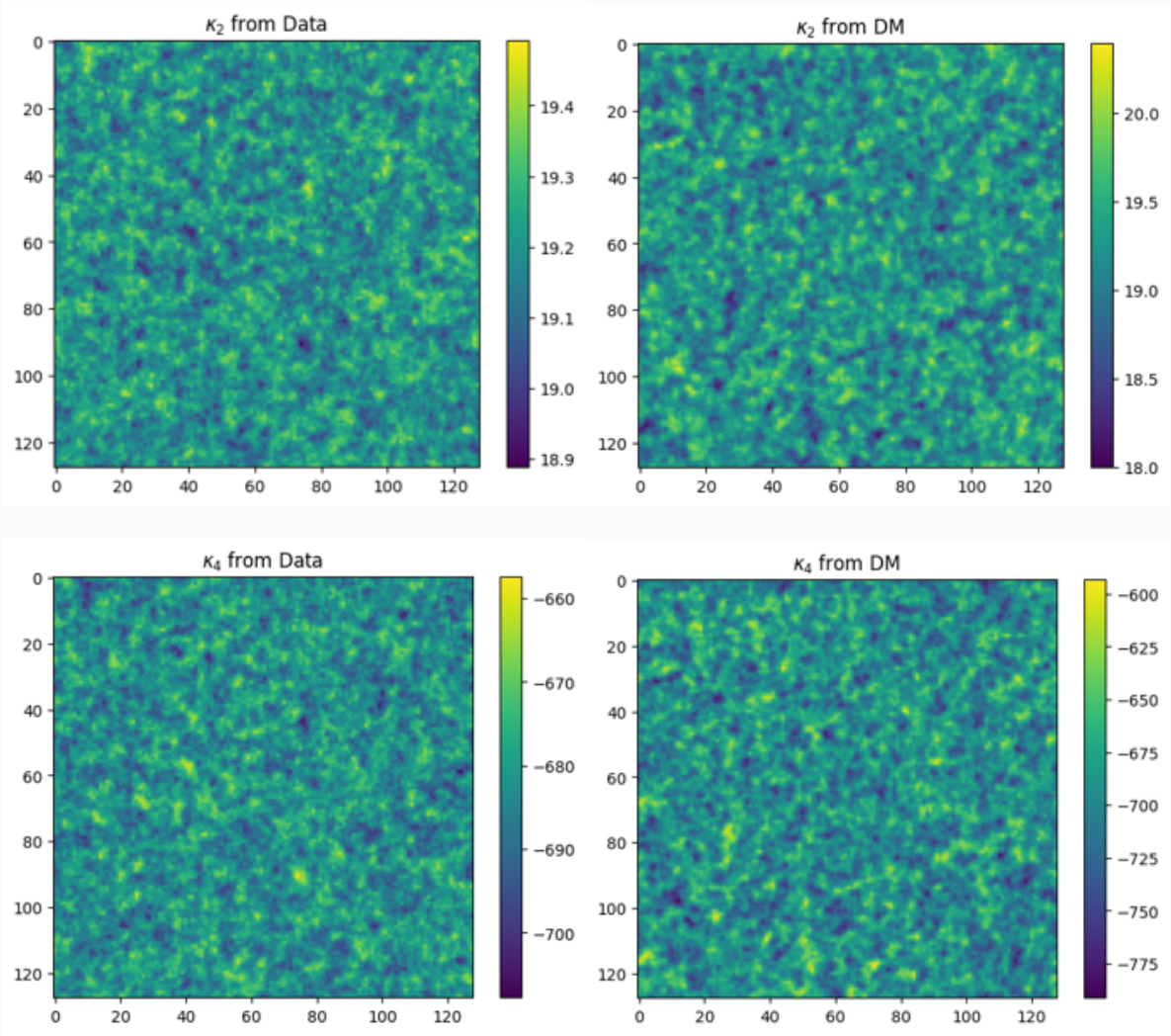
DM for 2D ϕ^4 Theory

Samples From the Data and DM



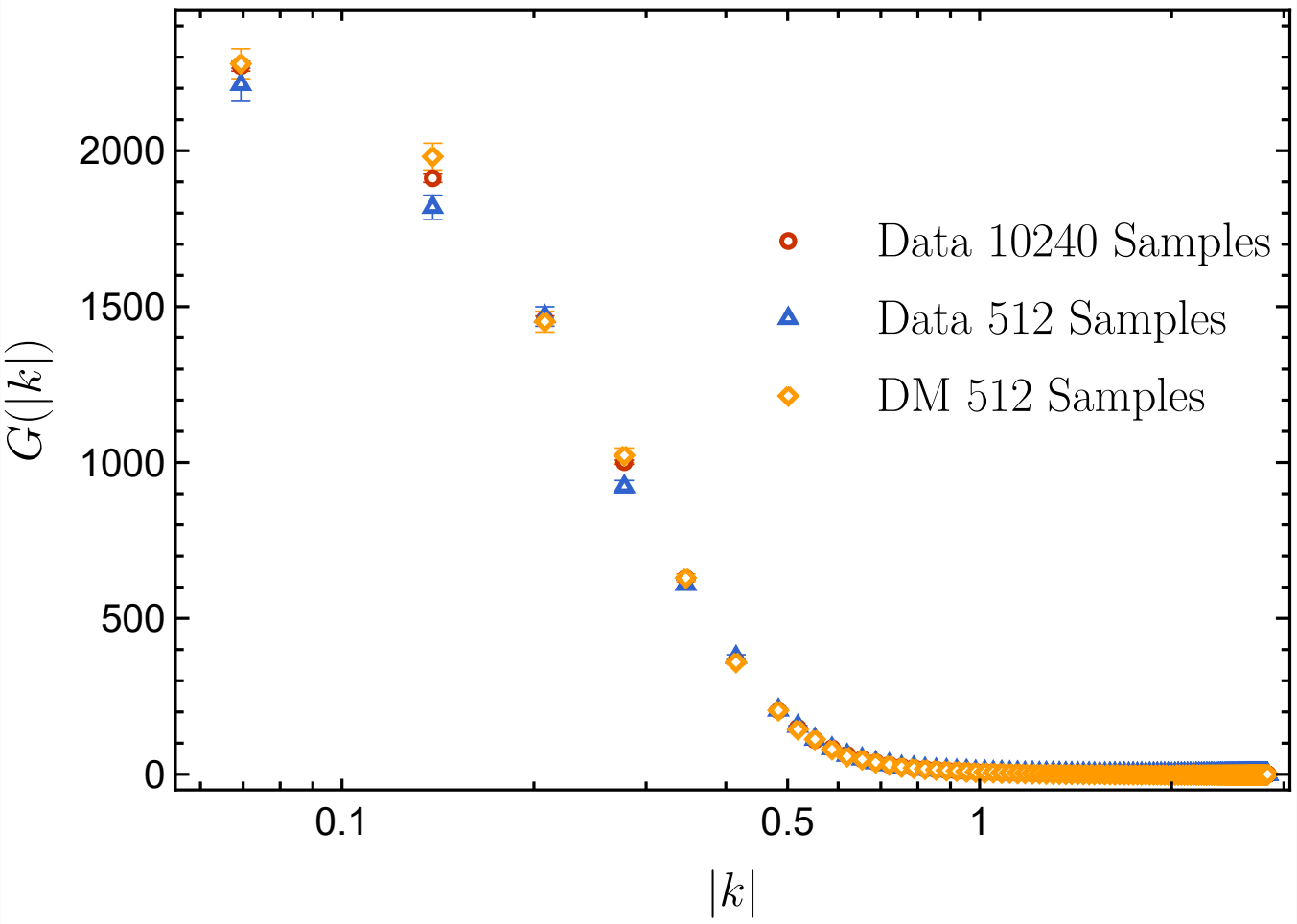
- Upper: Samples from the data
- Lower: Samples from the DM

Spatial distribution of Cumluant from the DM

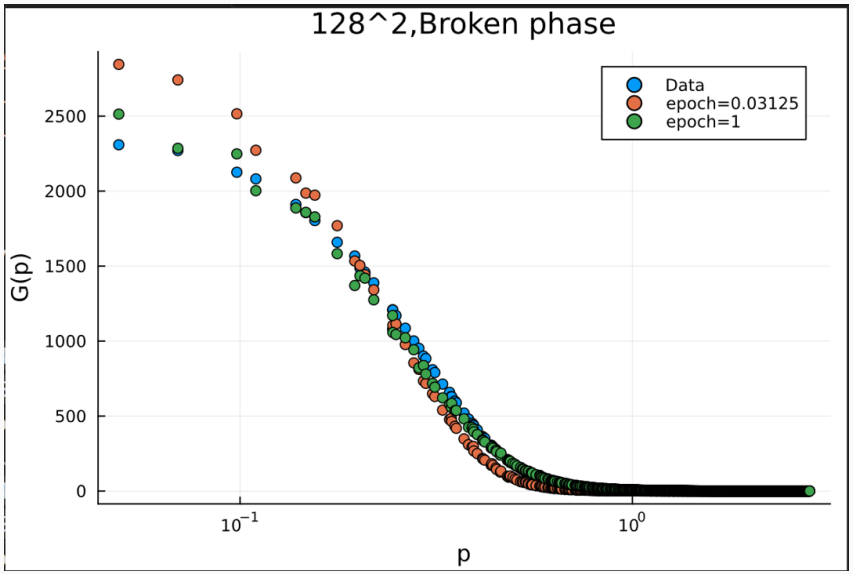


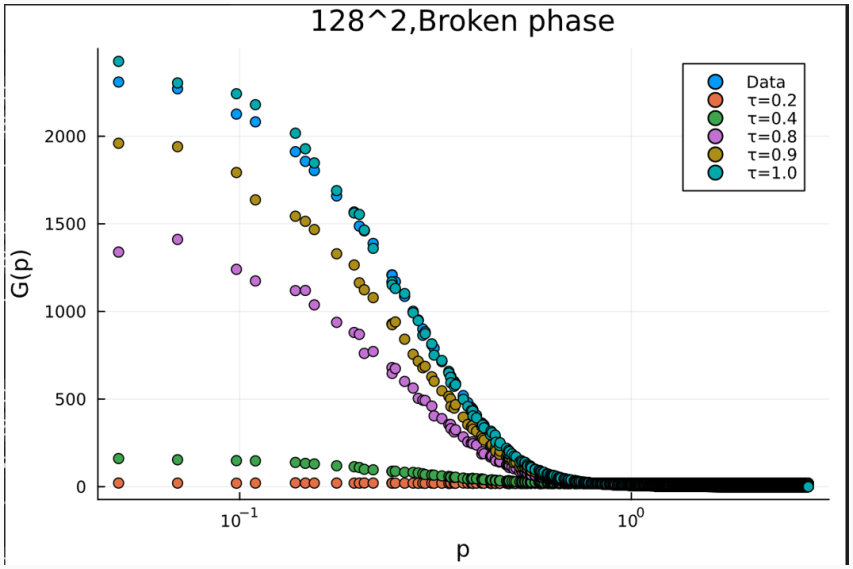
- Upper: κ_2
- Lower: κ_4

The propagator of the DM

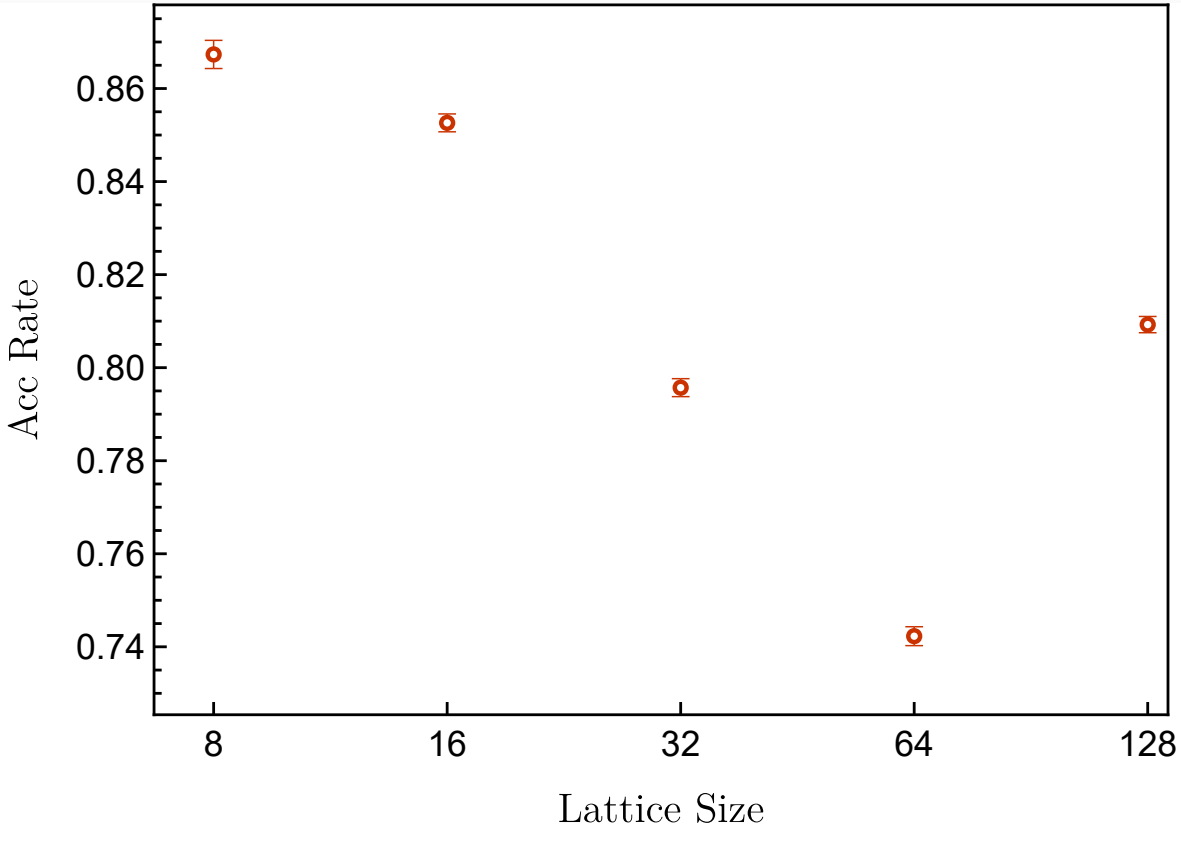


The evolution of the propagator during training





Acceptance Rate of the DM



Dissipative dynamics in score network

The reverse process of the score network can be written as:

$$\frac{\partial \phi(x,t)}{\partial t} = -g^2(t)\nabla_\phi \log p_t(\phi) + g(t)\eta(t,x) \tag{4}$$

where $g(t)$ is the noise scale and $\langle \eta(t,x)\eta(t',x')\rangle = \delta(t-t')\delta(x-x')$. and we use $s_\theta(\phi,t) = \nabla_\phi \log p_t(\phi)$ to approximate the score function.

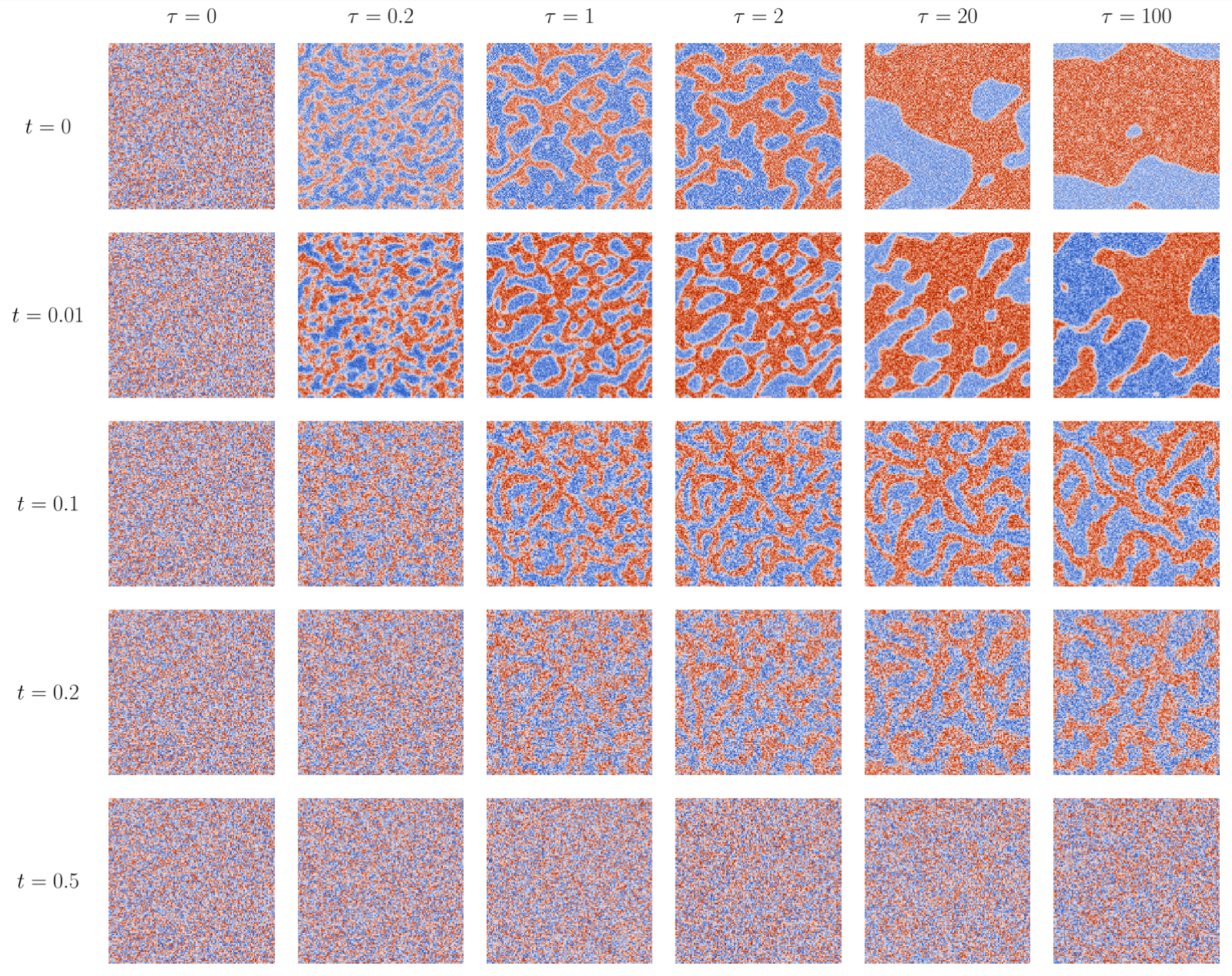
At a fixed time t, if we evolve the field $\phi(x)$ with Langevin dynamics as

$$\frac{\partial \phi(x,\tau)}{\partial \tau} = -g^2(t)s_\theta(\phi,t) + g(t)\xi(\tau,x) \tag{5}$$

then the field will converge to the distribution $p_t(\phi)$. The noise term $\langle \xi(\tau,x)\xi(\tau',x')\rangle = 2\delta(\tau-\tau')\delta(x-x')$.

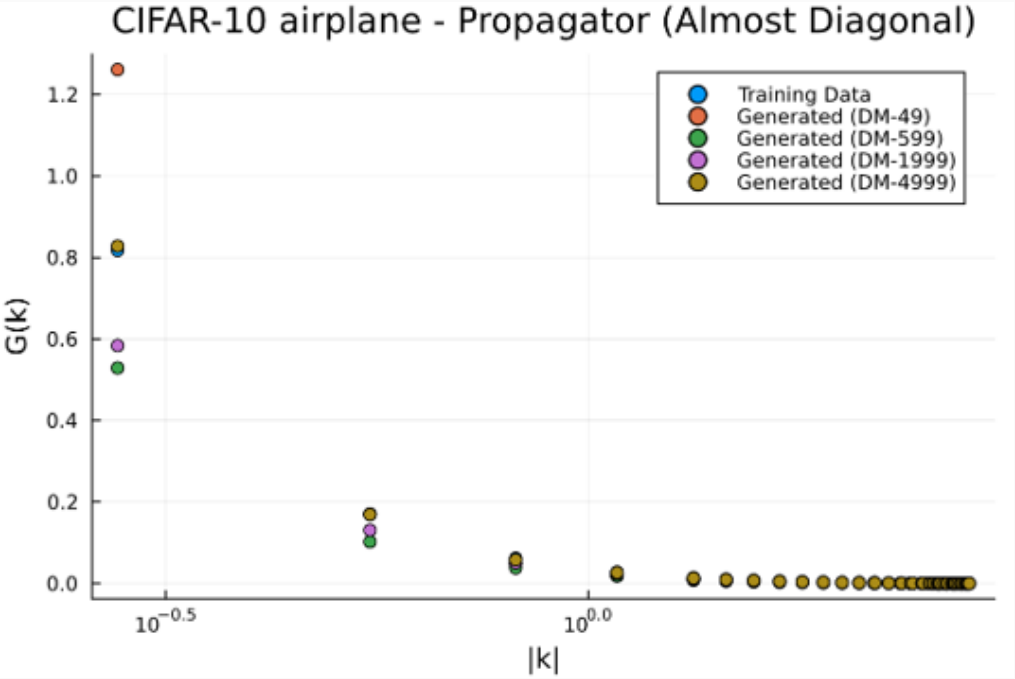
We can view $s_\theta(\phi,t) \approx \frac{\delta S_{\text{eff}}(\phi,t)}{\delta\phi}$ as the effective action.

Dissipative dynamics in score network

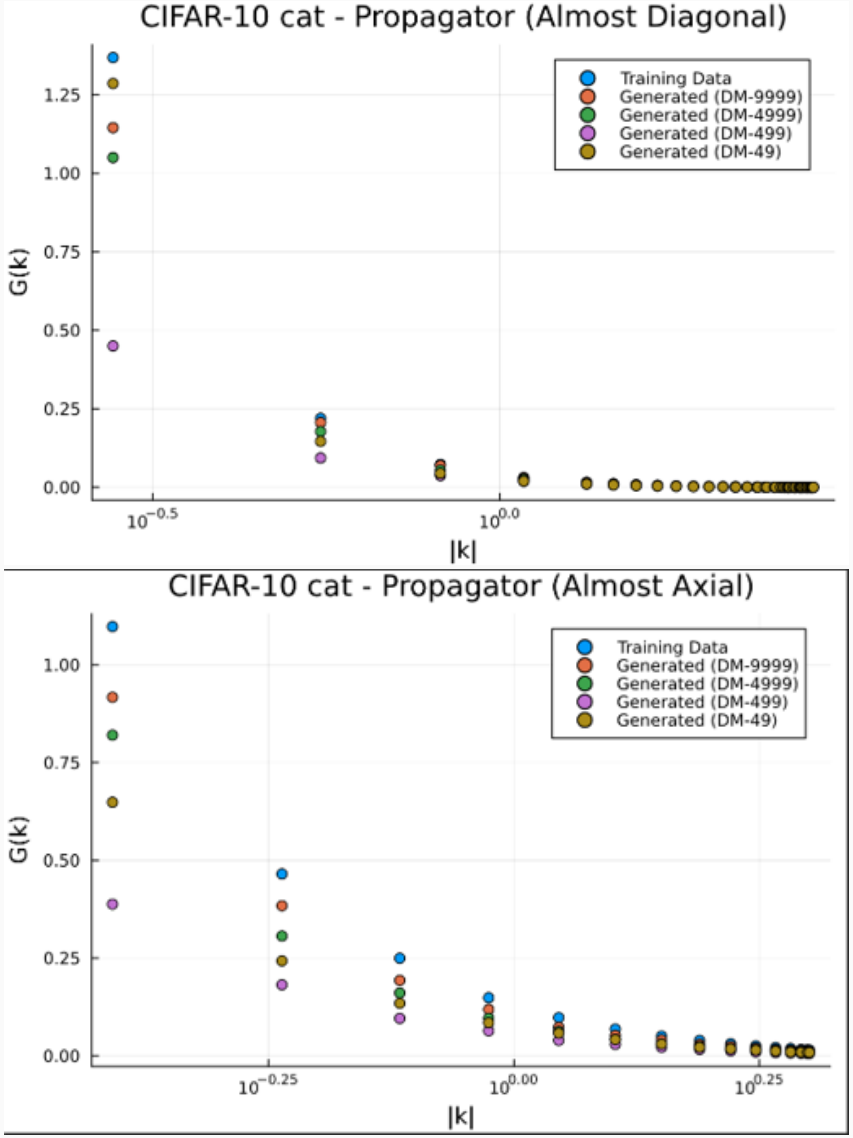


DM for Real Word Data

Porpagators from cifar10



Porpagators from cifar10



Backup

Determining σ_{max}

The noise schedule requires σ_{max} large enough to cover the data distribution.

Step 1: Normalize data to $[-1, 1]$:

$$\phi_{\text{norm}} = 2\left(\frac{\phi - \phi_{\text{min}}}{\phi_{\text{max}} - \phi_{\text{min}}} - 0.5\right) \tag{6}$$

Step 2: Compute pairwise Euclidean distances between samples:

$$d_{ij} = \|\phi_{\text{norm}}^{(i)} - \phi_{\text{norm}}^{(j)}\|_2 \tag{7}$$

Step 3: Set $\sigma_{\text{max}} \geq \max_{i,j} d_{ij}$ to ensure full coverage.

For 32^3 lattice with 1000 samples: $d_{\text{max}} \approx 204$, so we choose $\sigma_{\text{max}} = 205$. The theoretical maximum distance is $\sqrt{4 \times 32^3} \approx 362$ for 32^3 lattice.